

for all $\xi \in L^2(M, \mu)$ with $\int_M \xi(x) d\mu(x) = 0$ and $s \in S \setminus \{e\}$, where π is the unitary representation $\pi : G \rightarrow U(L^2(M, \mu))$ induced by the measure preserving action. Instead of considering the entire warped cone, he considered the coarse disjoint union of sections of $\mathcal{O}_G M$ at $t(n)$'s, where $\{t(n)\}_{n \in \mathbb{N}}$ is an increasing sequence of positive reals converging to infinity and we call it a warped system. Then he showed that an action has spectral gap if and only if for any (equivalently, some) sequence $\{t(n)\}_{n \in \mathbb{N}}$, the corresponding warped system is quasi-isometric to an expander.

In [WY12, WY14], Willett and Yu introduced a coarse invariance for metric spaces, so called geometric property (T), which is an obstruction to the surjectivity of the maximal coarse Baum-Connes assembly map. Roughly speaking, it is the existence of spectral gap for the Laplacian in the maximal Roe algebra associated with the space. So, for a coarse disjoint union of finite graphs, it is stronger than being an expander in the sense that the Laplacian has spectral gap in the Roe algebra. They showed that for a box space X of a group G , X has geometric property (T) if and only if G has property (T).

In [WY14], geometric property (T) was defined for discrete metric spaces, but to study it for warped systems, we adopt the non-discrete analogue formulated in [Win21]. If the base space is a compact Riemannian manifold M and the action is Lipschitz, the associated warped cone can be approximated by a graph with uniformly bounded degree as Lemma 2.7 and Lemma 2.8. Geometric property (T) in the sense of [Win21] is equivalent to the geometric property (T) of its discretization in the sense of [WY14]. In [Win21, Question 11.2], Winkel asked the following question:

Question 1.1. *If G has property (T), M is a compact Riemannian manifold and $G \curvearrowright M$ is an ergodic action by diffeomorphisms, then does the warped system has geometric property (T)?*

In this paper, we provide a negative answer to the question. The following is the main result.

Theorem 1.2. *Let G be a finitely generated group, and M a compact Riemannian manifold with the Riemannian measure μ and a G -action by homeomorphisms. Assume that $G \curvearrowright M$ is an ergodic, isometric and free action and μ is G -invariant. Then, for any sequence $\{t(n)\}_{n \in \mathbb{N}}$ converging to infinity, the associated warped system $\bigsqcup M \times \{t(n)\}$ does not have geometric property (T).*

While numerous studies, such as those in [Roe05, DN19, Saw19], have explored the similarities between the large-scale geometric properties of box spaces and warped cones, the above result shows that their behavior can differ significantly under maximal representations